Mr Dimelow

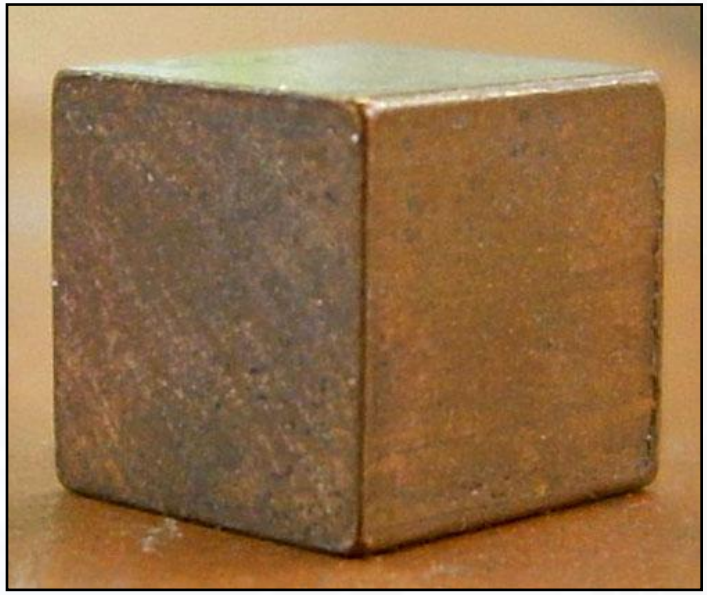
3.1 Engineering materials

3.1.1.1 Metals and alloys

Subject content	Additional information	Mathematical understanding
Ferrous metals and alloys: • cast iron • low and high carbon steels • steel alloys (stainless steel). Non-ferrous metals and alloys: • aluminium • copper • lead • zinc • alloys (brass and bronze).	Students will not be expected to have practical experience of working with all of these metals/alloys but exam questions could refer to any of the properties listed. Students should also be able to demonstrate knowledge and understanding of how the mechanical properties of these metals can change through: • the addition of materials to form alloys • methods which affect the grain size (heating) • cold working • hardening and quenching • corrosion • addition/subtraction of carbon in steels.	

Non-Ferrous Metals







In this presentation you will:

- recognise the range and characteristics of non-ferrous metals

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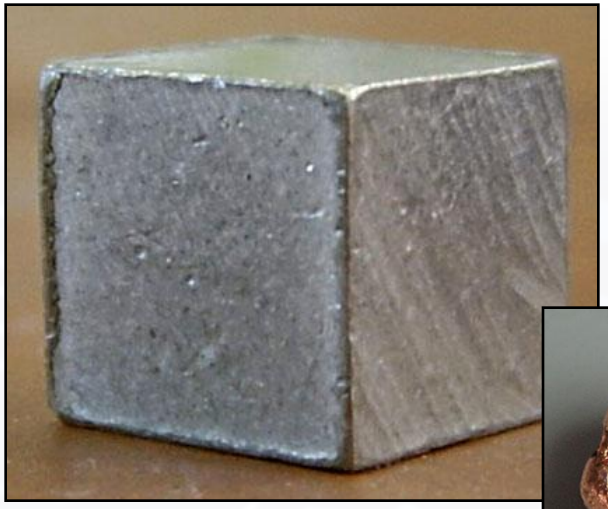
Introduction

Non-ferrous metals are metals that ***<u>do not</u>*** contain ***<u>iron</u>***.

Non-ferrous metals are ***<u>more expensive than</u>*** *ferrous* metals but are used for their *unique properties*.

Non-ferrous metals used in engineering include:

- ***Copper***
- ***Aluminium***
- ***Titanium***
- ***Refractory metals***







Aluminium

Aluminium is a ***silvery-white*** ***non-ferrous*** metal.



Aluminium has a ***low density***, just a ***third*** of the ***density of steel***. *While not as strong as steel, its* ***strength to weight ratio*** *is* ***superior***.

Aluminium is usually ***alloyed*** to ***improve its strength***. Its *strength is also improved by* ***heat treatment***.



Aluminium is ***alloyed*** with:

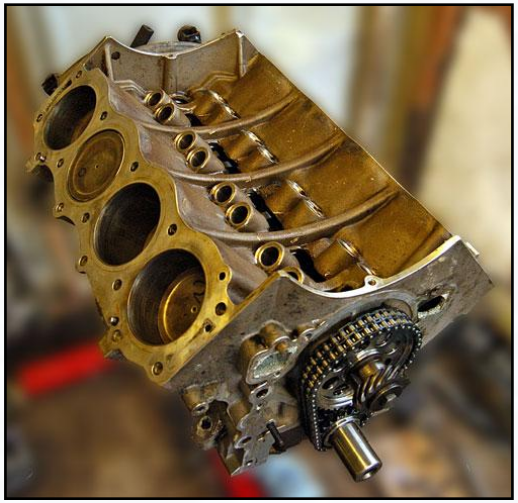
- ***Copper***
- ***Magnesium***
- ***Silicon***
- ***Manganese***

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Aluminium









Aluminium has many useful applications that make use of its properties, including ***high specific strength*** and ***electrical conductivity***.

Specific strength is the ***strength of the material per unit of mass***.

Copper

Copper is a pinkish brown non-ferrous metal.

Copper has been used for thousands of years in both its pure and alloyed form.



Pure copper


Ancient bronze

Copper can easily be ***strengthened*** by *alloying, heat treatment or work hardening*.

Bronze, an ***alloy*** of ***copper*** and ***tin***, was extremely important in ancient times.

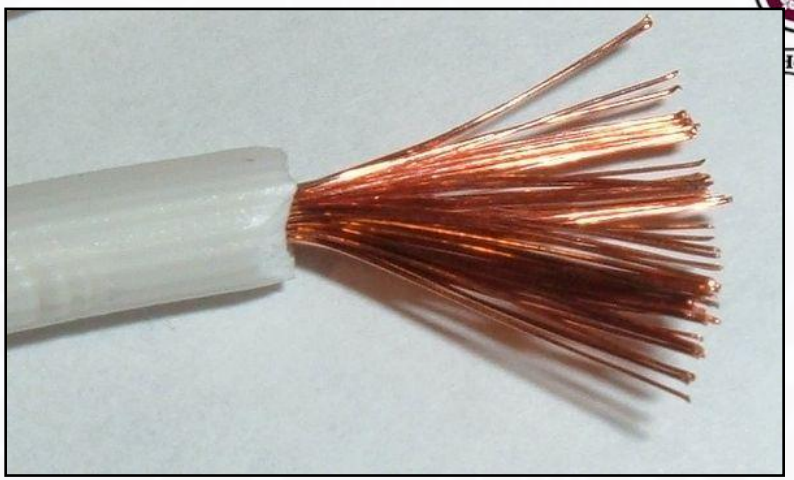
Brass is an ***alloy*** of ***copper*** and ***zinc***.


Brass gears

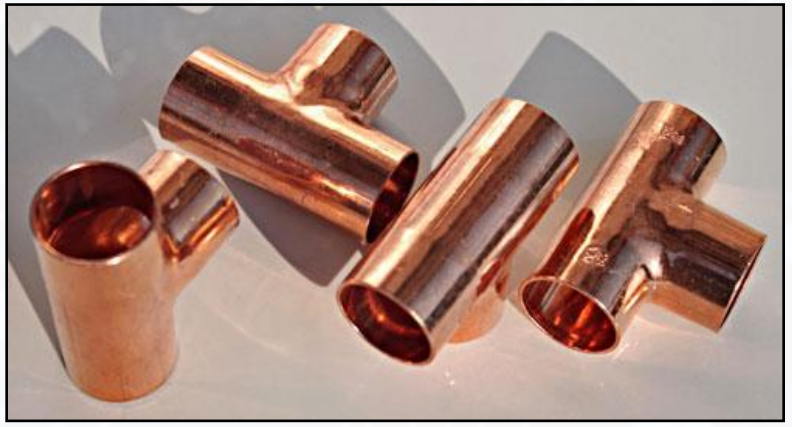
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Copper









Copper's most useful properties are its ***corrosion resistance*** and its ***extremely good electrical*** and ***thermal conductivity***.

Copper has many applications that make use of these properties.

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Alloys

Metals are not usually used in their pure form.

They are often combined with small amounts of other metals or elements to form a mixture called an alloy.

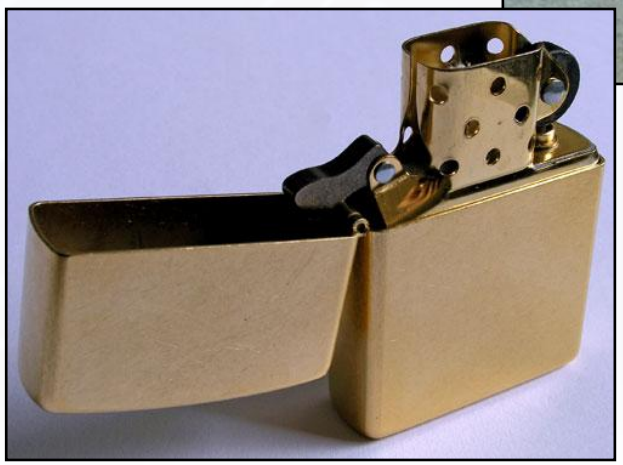
Generally, alloys have a better range of characteristics than pure metals.



Solder is an alloy of tin and lead



Stainless Steel is an alloy of iron, carbon and chromium



Brass is an alloy of copper and zinc

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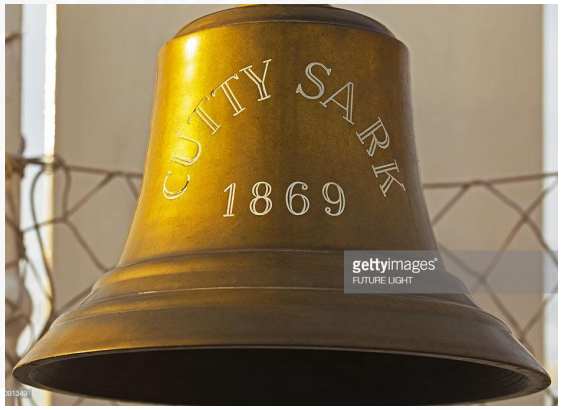
Brass (Copper Alloy)











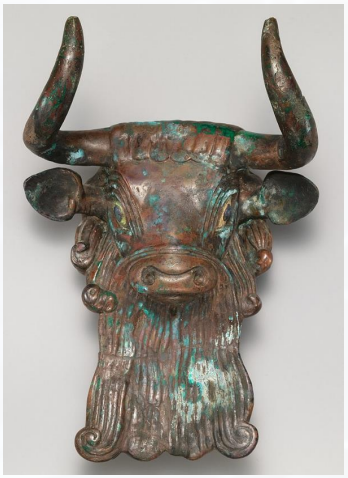
Brass is ***yellow/gold*** coloured ***metal***.

Brass *is the generic term for a range of* ***copper-zinc alloys*** *with differing combinations of properties, including strength, machinability, ductility, wear-resistance, hardness, colour, electrical and thermal conductivity, and corrosion resistance.*

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Bronze (Copper Alloy)









Bronze is ***reddish brown*** coloured ***metal***.

Bronze *is the generic term for a range of* ***copper-tin alloys*** *with differing combinations of properties, including strength, toughness, wear-resistance, machinability and corrosion resistance. Casts well.*

The tin is sometimes replaced with other metals but the alloy is still referred to as 'bronze'.

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Lead



Lead is a ***dull gray non-ferrous*** metal.

Lead is *relatively* ***soft, malleable*** *and* ***ductile*** *and has very good* ***resistance to corrosion***.



For these reasons lead sheets have often been used in the construction industry, e.g. to prevent water leaks around the edges of roofs.

It also has a ***high density***, so it is used for applications such as weights for diving belts and shielding for radiation in nuclear reactors.

It is a major constituent of some important ***alloys***:

- ***Solder – Lead/Tin***
- ***Pewter – Lead/Tin***



It used to be commonly used for water pipes in houses, but it was found that in the long term, exposure to lead can cause health problems in humans – less hazardous materials e.g. copper and uPVC are used for water pipes.

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Tin

Tin is a ***silvery-white non-ferrous*** metal.

Tin is a ***malleable*** metal with a ***low melting point***. One of its major uses is in ***solder***.

Tin is ***not easily oxidised***, so it is *often used to protect other metals from corrosion*.

It is a major constituent of some important ***alloys***:

- ***Bronze – Copper/Tin***
- ***Solder – Lead/Tin***
- ***Pewter – Lead/Tin***



Solder



A tin plated can

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Zinc

Zinc is a ***bluish-white*** ***non-ferrous*** metal.



Zinc is a ***hard, brittle*** metal and is widely used in various alloys.

Zinc is used as an ***anti-corrosion coating for steels***. The zinc is more reactive and protects the steel by corroding first. This is called ***galvanising***.



Zinc-plated roof



Galvanised steel

Zinc is commonly ***alloyed*** with:

- ***Copper***
- ***Aluminium***
- ***Nickel***

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Non-Ferrous Metals and their Alloys – *Properties and Uses*

Name and melting point	Composition	Properties and working characteristics	Uses
Aluminium 660°C	Pure metal	High strength/weight ratio, light, soft and ductile (FCC), work hardens in cold state, annealing necessary, difficult to join, non-toxic, good conductor of heat and electricity, corrosion-resistant, polishes well	Kitchen cooking utensils (pans) Packaging, cans, foils Window frames
Casting alloy (LM 4) (LM 6)	3% copper 5% silicon 12% silicon	Casts well, sand and die casting, good machineability, tougher and harder, increased fluidity	Engine components, cylinder heads
Duralumin	4% copper 1% manganese + magnesium	Almost the strength of mild steel but only 30% of the weight, hardens with age, machines well after annealing	Aircraft structure
Copper (Cu) 1083°C	Pure metal	Malleable, ductile (FCC), tough, suitable for hot and cold working, good conductor for heat and electricity, corrosion-resistant, easily joined, solders and brazes well, polishes well, rather expensive	Hot water storage cylinders Central heating pipes/tubing Wire electrical Copper clad board (circuits)
Copper alloys Gilding metal	15% zinc	Stronger, golden colour, enamels, easily-joined	Architectural metalwork Jewellery
Brass 900–1000°C	35% zinc	Corrosion-resistant, increased hardness, casts well, work hardens, easily joined, good conductor of heat and electricity, polishes well	Casting (e.g. valves) Boat fittings Ornaments
Bronze 900–1000°C	10% tin	Strong and tough, good wearing qualities, corrosion-resistant	Statues Coins Bearings
Tin (Sn) 232°C	Pure metal	Soft and weak, ductile and malleable, excellent resistance to corrosion even when damp, low melting point	Bearing metals Solder
Tin-plate	Steel plate tin coated	Bends with mild steel core, non-toxic	Tin cans
Lead (Pb) 327°C	Pure metal	Very heavy, soft, malleable and ductile but weak, corrosion-resistant, even by acid, low melting point, casts well, electrical properties	Roof coverings – flashings Plumbing Insulation against radiation
Zinc (Zn) 419°C	Pure metal	Very weak, poor strength/weight ratio, extremely resistant to atmospheric corrosion, low melting point, ductile (CPH) but difficult to work, expensive	Galvanised steel, dustbins Corrugated iron sheet roof Die casting alloys and rust proof paints

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Questions:

Answer the questions below. Illustrate your answers with diagrams or images were possible.

1. Explain the difference between ‘ferrous’ metals and ‘non-ferrous’ metals. Give two examples of ‘ferrous’ metals and two examples of ‘non-ferrous’ metals.
2. Explain the term ‘alloy' and give examples of two ‘ferrous’ and two ‘non-ferrous’ alloys.
3. List the benefits of using brass for the ‘pins’ on a 3-pin domestic plug. Why would copper be less suitable?
4. Define the materials that are used in the manufacture of a carving knife and a cold chisel. Explain how the properties and composition of each material contribute to the function of the product.

Questions:

Answer the questions below. Illustrate your answers with diagrams or images were possible.

Suggest suitable ***'metals'*** for each of the applications listed below. *Explain what material properties make it suitable.*

1. Electrical wire?
2. The machine bed of a centre lathe?
3. A twist drill or milling cutter?
4. A bolt used to secure the structural members of a bridge/building?
5. The engine block in a modern motor vehicle?
6. The fuselage of an aircraft?
7. A hip replacement?

Questions:

Answer the questions below. Illustrate your answers with diagrams or images were possible.

As design engineers it is essential that you understand what form the material you are specifying comes in.

Ferrous, non-ferrous and alloys are commonly available in what metal forms? For example a ‘sheet’ material.

Question 1

Which of the following is an example of a non-ferrous metal?

A) Iron

B) Steel

C) Stainless Steel

D) Titanium

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Question 1

Which of the following is an example of a non-ferrous metal?

A) Iron

B) Steel

C) Stainless Steel



D) Titanium

Non-ferrous metals do not contain iron.
Steel, iron and stainless steel all contain iron.

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Question 2

Which of the following metals is classed as a refractory metal?

A) Aluminium

B) Copper

C) Tungsten

D) Tin

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Question 2

Which of the following metals is classed as a refractory metal?

A) Aluminium

B) Copper



C) Tungsten

D) Tin

Refractory metals have high density and a high melting point.

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Question 3

Bronze is an alloy consisting mostly of which two metals?

A) Copper and tin

B) Copper and zinc

C) Aluminium and silicon

D) Tin and lead

Next >

Question 3

Bronze is an alloy consisting mostly of which two metals?



A) Copper and tin

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Question 4

Which of the following applications of aluminium alloys utilises its high strength to weight ratio?

A) Aircraft structures

B) Jet engine turbine blades

C) Anti-corrosion coatings

D) Coinage

Next >

Question 4

Which of the following applications of aluminium alloys utilises its high strength to weight ratio?



A) Aircraft structures

B) Jet engine turbine blades

C) Anti-corrosion coatings

D) Coinage

The good strength to weight ratio of aluminium makes it a valuable aerospace material.

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Summary

In this presentation you have seen:

- a range of non-ferrous metals and their characteristics and applications
- non-ferrous metals used as alloys

End